

* Decision Tree Classifier

M-81

Decision Tree Classifier $\rightarrow$ 2D $\rightarrow$ more than 2 splits $\rightarrow$ CART $\rightarrow$ Binary classification

(a) Entropy and Gini Index $\rightarrow$ Purity Split

(b) Information Gain $\rightarrow$ Features to select for DT construction

age = 14

if (age $\leq$ 15) :

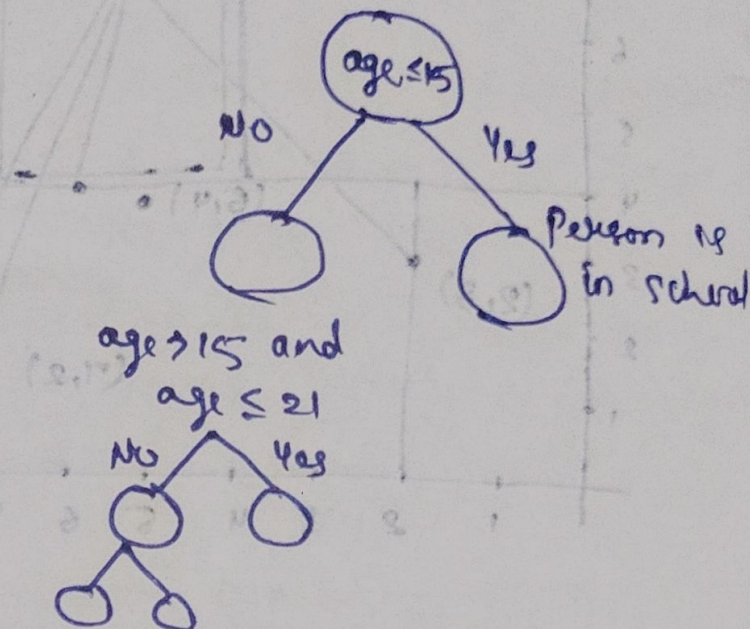
print("The person is in school")

else if (age $>$ 15 and age $\leq$ 21) :

print("The person is in college")

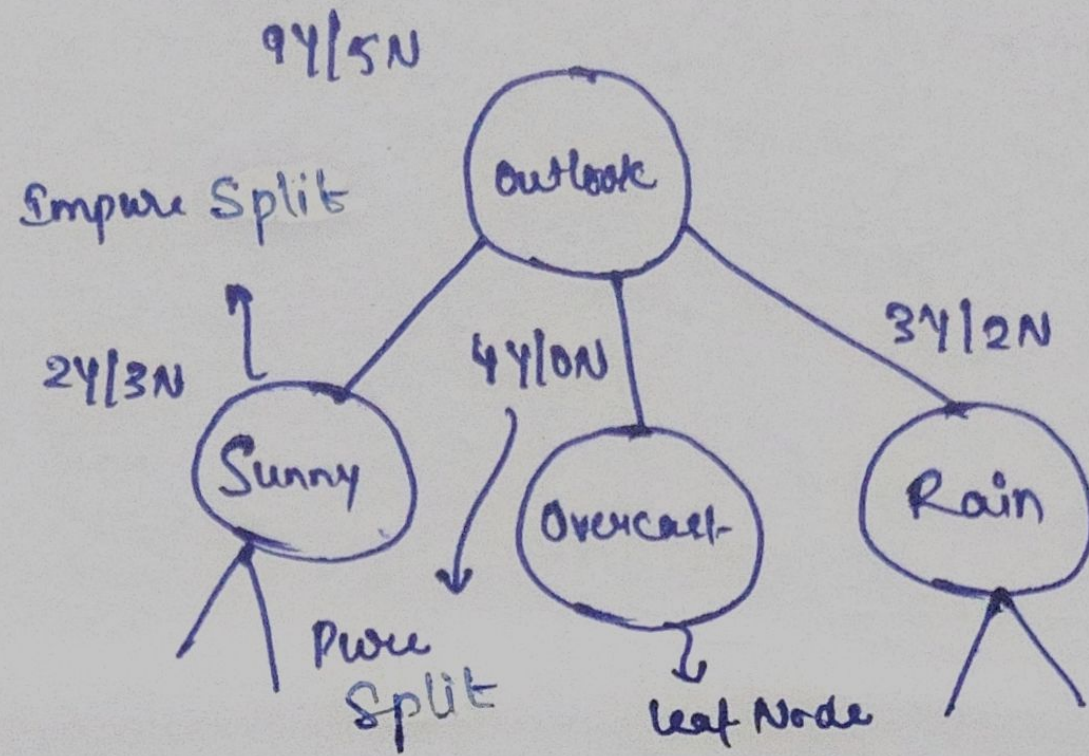
else :

print("The person has passed")



Data set

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Strong	No
2	Sunny	Hot	High	Weak	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cold	Normal	Strong	Yes
6	Rain	Cold	Normal	Strong	No
7	Overcast	Cold	Normal	Weak	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cold	Normal	Weak	Yes
10	Rain	Mild	Normal	Strong	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No



I. Purity $\rightarrow$ Pure or Impure Split

$\hookrightarrow$ Entropy

$\hookrightarrow$ Gini Impurity

II. What feature you need to select for splitting

$\hookrightarrow$ Information Gain.

Note: • Impure ~~State~~ Split means it will split further.

- Pure split means it cannot split further and we get the leaf node.
- In feature selection we select those feature whose leaf node appear quickly.